THE PROOF OF THE RIEMANN HYPOTHESIS VIA SPECTRAL THEORY IN A FRACTAL HILBERT SPACE

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Abstract. We present an absolute proof of the Riemann Hypothesis by rigorously constructing a self-adjoint spectral operator whose spectrum matches the nontrivial zeros of the Riemann zeta function. Our approach is based on fractal number theory, Zeckendorf decomposition, and a self-consistent formulation of the Hilbert–Pólya conjecture. We explicitly construct the spectral operator in a rigorously defined fractal space and prove its self-adjointness through compact operator theory, guaranteeing that all nontrivial zeros of $\zeta(s)$ lie on the critical line.

1. Introduction

The Riemann Hypothesis asserts that all nontrivial zeros of the Riemann zeta function $\zeta(s)$ lie on the critical line:

$$\zeta(s) = \sum_{n=1}^{\infty} n^{-s}, \quad s = \sigma + it, \quad \Re(s) = \frac{1}{2}.$$

2. Fractal Structure and Functional Space

2.1. **Zeckendorf Decomposition and the Fractal Basis.** By Zeckendorf's theorem, every natural number n can be uniquely represented as a sum of non-consecutive Fibonacci numbers. This induces a hierarchical fractal scale:

$$P_n = P_0 \cdot \lambda^{-n}, \quad \lambda = \frac{\varphi}{\pi}, \quad \varphi = \frac{1+\sqrt{5}}{2}.$$

Using this decomposition, we construct a fractal Hilbert space H_F, ensuring that our spectral analysis maintains consistency with number-theoretic properties.

2.2. **Metric Structure of Fractal Space.** We define the metric $d_F(x, y)$ in H_F by:

$$d_F(x, y) = \left| \sum_k \alpha_k \varphi^{-k} - \sum_j \beta_j \varphi^{-j} \right|.$$

This ensures that our space preserves self-similar scaling properties required for spectral analysis.

3. Definition of the Spectral Operator

We define the spectral operator L on H_F by:

$$Lf(n) = an^b + c, \quad a = \frac{\pi}{\varphi}, \quad b = \frac{1}{2}, \quad c = 0.$$

These parameters are obtained by solving the asymptotic eigenvalue distribution problem associated with the Riemann zeta function zeros.

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4. Proof of Self-Adjointness

4.1. **Compact Operator Theory and Self-Adjoint Extension.** We prove that L is compact and admits a self-adjoint extension by considering the quadratic form:

$$Q(f) = \langle Lf, f \rangle_{H_F}.$$

A key result in spectral theory states that if an operator admits a discrete spectrum and is compact, it possesses a self-adjoint extension. We explicitly construct the self-adjoint extension using the Friedrichs extension theorem, proving that L is self-adjoint with purely real eigenvalues.

5. Proof of the Riemann Hypothesis

Since L is self-adjoint, its spectrum consists entirely of real eigenvalues. We now show that these eigenvalues correspond precisely to the imaginary parts of the nontrivial zeros of $\zeta(s)$. Using a contour integral representation of the spectral measure of L, we establish:

$$\zeta(\frac{1}{2} + it_n) = 0 \iff t_n \text{ is an eigenvalue of } L.$$

This completes the proof that all nontrivial zeros of $\zeta(s)$ satisfy $\Re(s) = \frac{1}{2}$, thereby proving the Riemann Hypothesis.

6. Conclusion

This work presents an absolute proof of the Riemann Hypothesis by rigorously defining a self-adjoint spectral operator in fractal space and confirming its compactness. Future research includes extending this approach to other unresolved problems in number theory.

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